

Participant: SOLAR-WIND

You are responsible for the **SOLAR-WIND** technology in the market.

In real life, a bid includes multiple energy and price segments. However, for simplicity's sake, in this game you will only offer one segment. Therefore, your bid for a specific round will consist of the following:

- The **price** per megawatt-hour (€/MWh) at which you are willing to sell your energy. The minimum value (floor) is 0 €/MWh. The maximum value (cap) is 180 €/MWh.
- The amount of **energy** you are offering (GWh). The minimum value is 0 GWh. The maximum is 20 GWh.

Take into consideration that:

1. Your unitary variable cost associated with fuel consumption is **0 €/MWh**.
2. You don't have CO₂ emissions associated with generation.
3. Solar-Wind generation can be adjusted downward (i.e. you can curtail the power output), but not upward (your generation is limited by the available resource). Therefore, accurate forecasts are important for setting your bids and reducing imbalances. If you sell more energy in the market (at day *D-1*) than you actually generate at day *D*, you will incur a penalty according to the energy imbalance. The penalty price is different for each round, and depends on the overall imbalance, considering deviations from all participants.

For example, if you sell 12 GWh in the market but you generate only 10 GWh, you create an imbalance of 2 GWh. If the market clearing price is 20 €/MWh:

- You earn $12 \times 20 = 240$ thousand € (market incomes).
- If the penalty price were 30 €/MWh, you would pay a penalty of $2 \times 30 = 60$ thousand €.
- Thus, overall, you earn $240 - 60 = 180$ thousand € (total incomes).